13/10/25 "Tower" $E[E[Y|X]] = E[Y]$

"Prediction" $E[Y|X] = \arg\min_g E[(Y-g(X))^2]$

Pf: "Tower"

$$E[Y] = \sum_x \sum_y y\, p_{X,Y}(x,y) = \sum_x p_X(x) E[Y|X=x] = E[E[Y|X]]$$

($p_{X,Y}(x,y) = p_X(x)\, p_{Y|X=x}(y)$; "LOTUS"; $p_X(x)$)

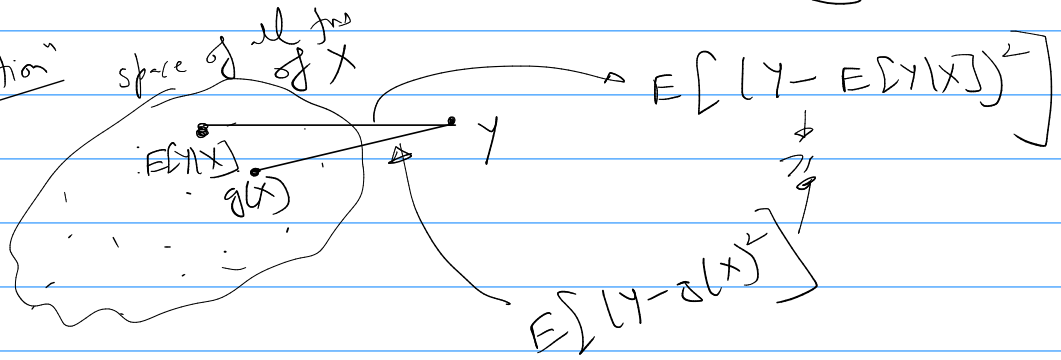
$$E[Y|X] \sim \{\ldots, E[Y|X=x], \ldots\}$$

"Tower modified"

$$E[Y h(X)] = \sum_x p_X(x) h(x) E[Y|X=x] = E[h(X) E[Y|X]]$$

$$E[Y h(X)] = E[E[Y h(X)|X]] = E[h(X) E[Y|X]]$$

"Prediction"



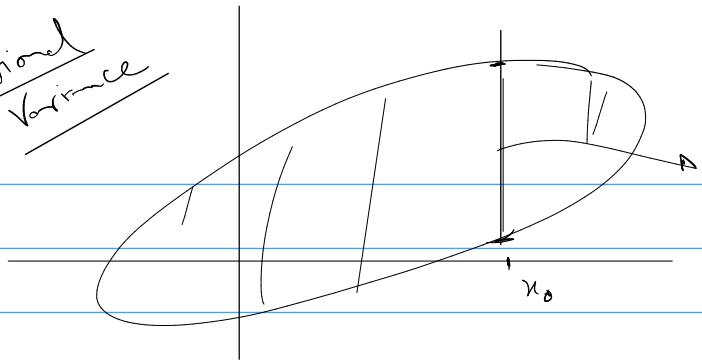
$$E[(Y - E[Y|X] + E[Y|X] - g(X))^2]$$

$$= E[(Y-E[Y|X])^2] + E[(E[Y|X]-g(X))^2] + 2E[(Y-E[Y|X])(E[Y|X]-g(X))]$$

($E[(E[Y|X]-g(X))^2] \geq 0$; $h(X)$; $(Y-E[Y|X])(E[Y|X]-g(X)) = 0$)

∎

Conditional Variance



$g(x_0) = E[Y|X=x_0]$

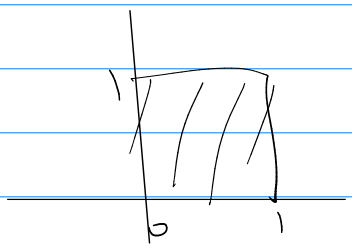
$E[Y|X] \triangleq g(X)$

$h(x_0) = Var(Y|X=x_0)$

$Var(Y|X) \triangleq h(X)$

Tower $Var(Y) = E[Var(Y|X)] + Var(E[Y|X])$

Ex: ①



$Y \sim Unif[0,1]$

$Y|X=x \sim Unif[0,1]$ → mean $1/2$, Var $1/12$

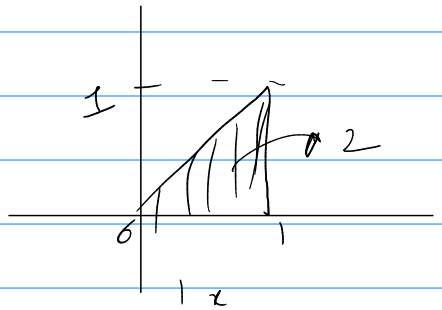
$E[Y|X] = 1/2$ w.p. 1

$Var(E[Y|X]) = 0$

$Var(Y|X) = 1/12$ w.p. 1

$\frac{1}{12} = Var(Y) = 1/12 + 0 = 1/12$

②



$E[X] = \int_{x=0}^{1}\int_{y=0}^{x} x \cdot 2 \, dy\,dx$

$= 2/3$

$E[X^2] = 1/2$

$Y|X=x \sim Unif[0,x]$

$E[Y|X] = \frac{X}{2}$ $\quad Var(E[Y|X]) = \frac{1}{4}Var(X)$

$= \frac{1}{4}\left(\frac{1}{2} - 4/9\right)$

$= 1/72$

$Var(Y|X) = \frac{X^2}{12}$

$Var(Y) = \frac{1}{24} + \frac{1}{72} = 1/18$

(3) $P \sim \text{Unif}[0,1]$

Given $P=p$, $X \sim \text{Binomial}(n,p)$

Predict P from X. $P|X=k$

$$f_{P|X=k}(p) = \frac{\Pr(X=k|P=p)\cdot f_P(p)}{\Pr(X=k)} = \frac{\binom{n}{k}p^k(1-p)^{n-k}\cdot 1}{\int_0^1 \binom{n}{k}p^k(1-p)^{n-k}\,dp}$$

$$\int_0^1 p^k(1-p)^l\,dp = \frac{k!\cdot l!}{(k+l+1)!}$$

$$= \frac{\binom{n}{k}p^k(1-p)^{n-k}}{\binom{n}{k}\frac{k!(n-k)!}{(n+1)!}} = \binom{n}{k}\cdot\frac{(n+1)!}{n!}p^k(1-p)^{n-k}$$

$p\in[0,1]$

$$E[X] = E[E[X|P]] = E[nP] = \frac{n}{2}$$

$$\text{Var}(X) = E[\underbrace{\text{Var}(X|P)}_{nP(1-P)}] + \text{Var}(\underbrace{E[X|P]}_{nP}) = \cdots$$

$$E[P|X=k] = \binom{n}{k}\cdot\frac{(n+1)!}{n!}\cdot\int_0^1 p^{k+1}(1-p)^{n-k}\,dp$$

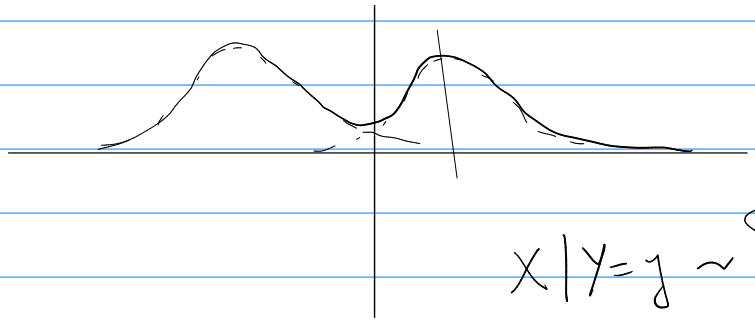
$$= \binom{n}{k}\frac{(n+1)!}{n!}\cdot\frac{(k+1)!(n-k)!}{(n+2)!} = \frac{k+1}{n+2}$$

$$E[P|X] = \frac{X+1}{n+2} \longrightarrow \text{"add-1" estimator}$$

$$\text{Var}(P|X) = ? \qquad \text{Var}(E[P|X]) = \frac{\text{Var}(X)}{(n+2)^2} = ?$$

(4) $X \sim \{-1, 1\}$ (each $1/2$)

$Y|X=x \sim \text{pdf } \frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(y-x)^2}{2\sigma^2}}$, μ_y



$$X|Y=y \sim \{\overset{?}{-1}, \overset{?}{1}\}$$

$$\Pr(X=1|Y=y) = \frac{f_{Y|X=1}(y)\cdot P(X=1)}{f_Y(y)}$$

$$= \frac{\frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(y-1)^2}{2\sigma^2}} \cdot \frac{1}{2}}{\frac{1}{\sqrt{2\pi}\sigma}\cdot\frac{1}{2}\left(e^{\frac{-(y-1)^2}{2\sigma^2}} + e^{\frac{-(y+1)^2}{2\sigma^2}}\right)}$$

$$E[X|Y=y] = \frac{e^{\frac{-(y-1)^2}{2\sigma^2}} - e^{\frac{-(y+1)^2}{2\sigma^2}}}{e^{\frac{-(y-1)^2}{2\sigma^2}} + e^{\frac{-(y+1)^2}{2\sigma^2}}} = \frac{e^{y/\sigma^2} - e^{-y/\sigma^2}}{e^{y/\sigma^2} + e^{-y/\sigma^2}}$$

$$= \tanh\left(y/\sigma^2\right)$$

